Rational Functions & Asymptotes

ANSWER KEY

Section 1: Vertical Asymptotes vs. Removable Discontinuities

1 Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 9}{x^2 - x - 6}$.

a. **Vertical asymptote at $x = -2$; hole at $x = 3$** ✓

b. Vertical asymptote at $x = 3$; hole at $x = -2$

c. Vertical asymptotes at $x = 3$ and $x = -2$; no holes

d. Holes at $x = 3$ and $x = -2$; no vertical asymptotes

Factor: $\frac{(x-3)(x+3)}{(x-3)(x+2)}$. The factor $(x - 3)$ cancels, indicating a removable discontinuity (hole) at $x = 3$. The remaining denominator factor $(x + 2)$ yields a vertical asymptote at $x = -2$.

2 What is the coordinate (x, y) of the hole in the graph of $g(x) = \frac{2x^2 + 6x}{x^2 - 9}$?

a. $(-3, 1)$ ✓

b. $(-3, 0)$

c. $(3, 1)$

d. $(-3, \infty)$

Factor: $\frac{2x(x+3)}{(x-3)(x+3)} = \frac{2x}{x-3}$ for $x \neq -3$. At $x = -3$, $y = \frac{2(-3)}{-3-3} = \frac{-6}{-6} = 1$. The hole is at $(-3, 1)$.

3 If $h(x) = \frac{x-4}{(x-4)^2(x+1)}$, which describes the behavior of h near $x = 4$?

a. $h(x)$ **has a vertical asymptote at** $x = 4$ ✓

b. $h(x)$ has a removable hole at $x = 4$

c. $h(x)$ has a horizontal tangent at $x = 4$

d. $h(x)$ is continuous at $x = 4$

Simplifying: $h(x) = \frac{1}{(x-4)(x+1)}$ for $x \neq 4$. Because $(x - 4)$ remains in the denominator to degree 1, $x = 4$ is a vertical asymptote, not a hole.

Section 2: Horizontal Asymptotes and Degrees

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4 Determine the horizontal asymptote of $R(x) = \frac{5x^3 - 2x + 1}{2x^3 + 7x^2 - 4}$.

a. $y = \frac{5}{2}$ ✓

b. $y = 0$

c. $y = -\frac{1}{4}$

d. No horizontal asymptote

The degrees of numerator and denominator are both 3. The horizontal asymptote is the ratio of leading coefficients: $y = \frac{5}{2}$.

5 Which rational function has a horizontal asymptote at $y = 0$?

a. $f(x) = \frac{4x + 1}{3x^2 - 5}$ ✓

b. $g(x) = \frac{3x^2 - 5}{4x + 1}$

c. $h(x) = \frac{4x^2 + 1}{3x^2 - 5}$

d. $k(x) = \frac{4x^3}{3x^2 - 5}$

A rational function has $y = 0$ as its horizontal asymptote when the degree of the denominator is strictly greater than the degree of the numerator. Here degree of numerator is 1 and denominator is 2.

6 Can the graph of a rational function intersect its horizontal asymptote?

a. **Yes, the graph can cross a horizontal asymptote at finite values of** x ✓

b. No, an asymptote can never be touched or crossed by definition

c. Only if the horizontal asymptote is $y = 0$

d. Only if the function has no vertical asymptotes

Horizontal asymptotes describe end behavior as $x \to \pm\infty$. The graph of a rational function can intersect a horizontal asymptote one or more times in the interior/finite domain.

Section 3: Slant (Oblique) Asymptotes

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7 Find the slant asymptote of $f(x) = \frac{2x^2 + 5x - 1}{x + 2}$.

a. $y = 2x + 1$ ✓

b. $y = 2x + 9$

c. $y = 2x$

d. $y = x + 1$

Perform polynomial long division: $(2x^2 + 5x - 1) \div (x + 2) = 2x + 1 + \frac{-3}{x+2}$. As $x \to \pm\infty$, $\frac{-3}{x+2} \to 0$. The slant asymptote is $y = 2x + 1$.

8 Under what condition does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. **The degree of $P(x)$ is exactly 1 greater than the degree of $Q(x)$** ✓

b. The degree of $P(x)$ equals the degree of $Q(x)$

c. The degree of $Q(x)$ is exactly 1 greater than the degree of $P(x)$

d. The degree of $P(x)$ is at least 2 greater than the degree of $Q(x)$

A slant asymptote exists if and only if the numerator's polynomial degree is exactly 1 higher than the denominator's polynomial degree.

9 Find the equation of the slant asymptote for $g(x) = \frac{x^3 - 3x^2 + 4}{x^2 + 1}$.

a. $y = x - 3$ ✓

b. $y = x + 3$

c. $y = x$

d. $y = -3x$

Dividing $x^3 - 3x^2 + 4$ by $x^2 + 1$: $x(x^2 + 1) = x^3 + x$, leaving remainder $-3x^2 - x + 4$. Next term: $-3(x^2 + 1) = -3x^2 - 3$, leaving remainder $-x + 7$. The quotient is $x - 3$, so the slant asymptote is $y = x - 3$.

Section 4: Limit Notation and Asymptotic Behavior

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10 For $f(x) = \frac{1}{x-3}$, what are $\lim_{x \to 3^+} f(x)$ and $\lim_{x \to 3^-} f(x)$?

a. $\lim_{x \to 3^+} f(x) = \infty$ **and** $\lim_{x \to 3^-} f(x) = -\infty$ ✓

b. $\lim_{x \to 3^+} f(x) = -\infty$ and $\lim_{x \to 3^-} f(x) = \infty$

c. Both limits equal ∞

d. Both limits equal $-\infty$

As $x \to 3$ from the right, $x - 3 > 0$, so $\frac{1}{x-3} \to +\infty$. As $x \to 3$ from the left, $x - 3 < 0$, so $\frac{1}{x-3} \to -\infty$.

11 If $\lim_{x \to -\infty} f(x) = 4$, which statement is justified?

a. **The line $y = 4$ is a horizontal asymptote of f** ✓

b. The line $x = 4$ is a vertical asymptote of f

c. $f(4)$ is undefined

d. The range of f is $[4, \infty)$

By definition, if $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$, the horizontal line $y = L$ is a horizontal asymptote.

12 Given $k(x) = \frac{2x-6}{(x-3)(x-5)}$, evaluate $\lim_{x \to 3} k(x)$.

a. -1 ✓

b. 1

c. ∞

d. 0

Factor and cancel: $k(x) = \frac{2(x-3)}{(x-3)(x-5)} = \frac{2}{x-5}$ for $x \neq 3$. Taking the limit as $x \to 3$: $\frac{2}{3-5} = \frac{2}{-2} = -1$.

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